

SHAIKH ZEESHAN MURSHED

Senior Frontend Engineer | React | TypeScript | Full-Stack

Montréal, QC, Canada | Open to Remote | sk.zeeshan1992@gmail.com | [linkedin.com/in/skizsk](https://www.linkedin.com/in/skizsk) | github.com/sk-izsk | izsk.dev | izsk.hashnode.dev

Languages: English, Bengali, Hindi

PROFESSIONAL SUMMARY

Senior Front-end React Developer with 7+ years building high-performance, accessible web interfaces for enterprise SaaS platforms. Deep expertise in React, TypeScript, and component architecture – from designing reusable UI libraries consumed across multiple teams to translating complex UX/UI mockups into maintainable, production-grade interfaces. Consistent track record of optimizing front-end performance, instrumenting analytics across user journeys, and integrating cleanly with REST APIs and backend teams. Recently extended into full-stack LLM integration: built and deployed a FastAPI backend powering a portfolio Ask AI feature, with Groq inference, Upstash Redis caching, IP-based rate limiting, and a structured prompt pipeline. Comfortable owning technical decisions end-to-end, contributing to agile ceremonies, and raising the quality bar through code review, documentation, and mentoring.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, HTML5, CSS3

Frameworks: React, Next.js, Node.js, Express, FastAPI, GraphQL, React Native (Expo)

State / Data: TanStack Query (React Query), Redux Toolkit, Zustand

UI Libraries: Material-UI (MUI), Ant Design, Tailwind CSS, Shadcn, NativeWind, Styled-Components, Bootstrap, JSS

Testing: Jest, Vitest, Cypress (E2E), React Testing Library, React Native Testing Library, Enzyme, Mocha, Chai, Nock

Tools / DevOps: Git, GitLab CI/CD, Docker, Webpack, Vite, Storybook, Postman, Figma, Jira, Notion, Bun

Visualization / 3D: Three.js, React Three Fiber, Drei, Mafes, Framer Motion

Real-Time: Socket.IO, WebSockets, Server-Sent Events (SSE)

Analytics / Monitoring: Frontend event tracking, user funnel instrumentation, error monitoring (Sentry), performance metrics (Core Web Vitals)

Databases: PostgreSQL, MongoDB, Drizzle ORM, Upstash Redis

Auth / Cloud: Clerk, JWT, AWS (S3, Lambda), Cloudinary, Heroku, Netlify, Render

AI & Agentic Tooling: Prompt engineering, LLM integration (Groq, llama-3.3-70b), FastAPI LLM backends, Upstash Redis caching, IP rate limiting, MCP (custom servers, Storybook, Figma, GitHub), agentic workflow automation, AI code review

EXPERIENCE

Software Engineer

Nov 2020 – Present

SwiftConnect — Montréal, QC

Front-end lead on enterprise access & security management SaaS platform

- Designed and evolved high-performance web interfaces using React, TypeScript, Styled-Components, Ant Design, Redux Toolkit, and TanStack Query, covering complex authorization workflows across multiple product areas.
- Translated UX/UI mockups into reusable, accessible component systems (WCAG standards) across 10+ UI components, cutting design-to-dev revision cycles by ~30%.
- Authored and maintained a private, Storybook-documented NPM component library adopted across cross-functional teams, reducing duplicate front-end code by ~40% and enforcing consistent UI standards at scale.
- Architected and published a private Node.js SDK (core JavaScript, internal NPM registry) wrapping all backend API calls behind a versioned, typed interface – eliminating duplicated HTTP logic across projects and distributed to third-party customers integrating SwiftConnect's API into their own systems without adopting the full platform frontend.
- Built a companion React SDK layered over the Node SDK, exposing React-specific hooks and components for API integration – adopted by external customers building custom frontends against SwiftConnect's access management API.
- Solely designed, built, and maintained an internal React Native (Expo) dashboard application for the customer-facing team, as the only engineer on the team with prior React Native experience – set up the project from scratch, including navigation/routing, API integration wired into the existing SDK and design system, data-heavy dashboard views, and full unit test coverage via React Native Testing Library, all while maintaining the app solo alongside other ongoing projects.
- Engineered a React-based email templating pipeline where developers author SendGrid templates as React components; merged changes trigger an automated CI/CD deployment to SendGrid, keeping all email output version-controlled and eliminating manual template management.
- Instrumented front-end analytics and KPI tracking across key user journeys, giving product teams real-time visibility into feature adoption, funnel drop-off, and performance metrics.

- Optimized front-end performance and code quality through GitLab CI/CD pipeline setup, Core Web Vitals tracking, and consistent test coverage via Jest and React Testing Library.
- Instrumented Sentry error monitoring across all front-end surfaces, configuring release tracking, source map uploads, and alert routing to reduce mean time to detection on production issues.
- Mentored 2–3 junior developers through code reviews, pair programming, and bi-weekly knowledge-sharing sessions, reducing review turnaround time by ~25%.
- Acted as technical lead and proxy Product Owner during critical delivery phases, translating business requirements into architecture plans and coordinating sprint execution across front-end and backend teams.
- Built a Storybook + Figma MCP pipeline for commit-time design-integrity validation, and engineered custom MCP servers enabling in-editor API doc queries, reducing cross-functional revision cycles by ~35%.
- Integrated AI tooling into engineering workflows: LLM-assisted PR reviews via GitHub MCP, agentic scaffolding for boilerplate and test generation, all with human-review gates.

Full Stack Developer

Oct 2020 – Nov 2020

Hatchways — Remote

Agile team of 4, full-stack web application development

- Built web applications using React and Material-UI on the front end and Express, Node.js, MongoDB, Cloudinary, and JWT authentication on the backend.
- Delivered full-stack features in 2-week sprint cycles within a cross-functional Agile team of 4.

Junior Software Developer

July 2019 – June 2020

Guiker Inc. — Montréal, QC

Mobile-first apartment rental marketplace startup

- Contributed to a full React rewrite of the main web product using TypeScript, Material-UI, RESTful APIs, and JSS with comprehensive Jest test coverage.
- Integrated REST API endpoints and owned specific Node.js/Express routing and middleware logic, directly reducing front-end-to-backend integration issues.
- Implemented in-app landlord-tenant chat and lease-signing flows, contributing to a 15–20% increase in repeat session activity.
- Developed mobile application features for iOS and Android using React Native (Expo), TypeScript, and Jest unit testing.

PROJECTS

SwiftConnect | Enterprise Access & Security Management Platform

- **Core Web Client:** Designed the primary access control interface in React/TypeScript. Built global state and caching layers (Redux Toolkit, TanStack Query) to manage complex authorization data, keep the UI stable during heavy mutations, and surface real-time analytics on feature adoption and conversion funnels.
- **Internal React Native Dashboard:** Solo-built an internal React Native (Expo) application for the customer team from scratch – project setup, routing/navigation, API integration layered on the existing Node/React SDK and design system, and data-heavy dashboard screens, all covered by unit tests with React Native Testing Library and maintained independently alongside other workstreams.
- **SDK Distribution:** Architected a Node.js SDK and companion React SDK (private NPM) exposing typed, versioned API interfaces – adopted internally across all projects and distributed externally to third-party customers building custom frontends against SwiftConnect's access management API.
- **Design System:** Authored a private, strongly-typed NPM component library enforced across cross-functional teams, reducing duplicate front-end code by ~40% and maintaining consistent UI standards across the platform.
- **Front-end Quality & Observability:** Established CI/CD pipelines, Sentry error monitoring (release tracking, source maps, alert routing), and Core Web Vitals performance tracking across all front-end surfaces.
- **AI-Powered Developer Experience:** Built custom MCP servers (Storybook, Figma, GitHub) to automate design validation, streamline PR workflows, and eliminate developer context-switching.

Portfolio Ask AI | Full-Stack LLM Feature – React Frontend + FastAPI Backend | izsk.dev | github.com/sk-izsk/zsk-ai-resume-api

- Built an end-to-end Ask AI feature across two repos: a React 19 / TypeScript / Vite frontend with a /ask-ai route, and a Python FastAPI backend deployed on Render, integrating Groq's llama-3.3-70b-versatile model for resume-grounded question answering.
- Designed the backend prompt pipeline to enforce strict answer grounding: professional experience cited first, portfolio projects second, skill ratings last – with a blocked response field for off-topic queries and JSON-only structured output (answer, sources, blocked).
- Implemented Upstash Redis REST-based caching with a versioned cache key (CACHE_VERSION) for prompt/behavior invalidation, and IP-based rate limiting (3 requests/IP/hour) with clean 429 error handling propagated through to the frontend.

- Built a `compact_profile()` pipeline that sends a reduced resume context to Groq instead of the full 54KB JSON, cutting token usage and keeping inference costs manageable.
- Frontend handles Render free-tier cold starts with a `/health` pre-flight call that blocks the chat panel with a loading overlay until the backend wakes, then fires `POST /chat` – surfacing backend-provided `detail.message` for rate limit errors.
- Chat UI features prompt buttons, message bubbles, assistant typing indicator, evidence/source dropdowns, auto-growing input capped at a max height, and themed scrollbars – with keyboard shortcuts (Enter to submit, Shift+Enter for newline) and remaining-request count display.

Guiker | *Rental Marketplace — Apartment Search, Lease Signing & Chat*

- **Core Web Re-Architecture:** Full rewrite of the legacy web platform into a strongly-typed React SPA with modular, reusable UI components using Material-UI and JSS.
- **REST API Integration:** Owned the Node.js/Express layer with custom endpoints, middleware validation, and controller logic, reducing front-end-to-backend integration bottlenecks.

Whisper Chat | *Full-Stack Real-Time Messaging App* | Live Demo | github.com/sk-izsk/react-native-fullstack-whisper-chat

- Built a real-time chat system across three surfaces: a React 19 / Vite web client, a React Native / Expo / Sentry mobile client, and a shared Express 5 / Socket.IO / MongoDB TypeScript backend, with Clerk authentication integrated consistently on all three.
- Split state across two layers: TanStack Query for server state (chats, messages, users) and Zustand for transient realtime state (socket connection, unread markers, typing indicators, online presence), keeping cache invalidation logic isolated from socket event handling.
- Refactored socket logic into dedicated modules (client, events, cache handlers, store) and split oversized chat screens into focused hooks and components; added optimistic message sending, Sentry error monitoring on mobile, and consistent product behavior across web and native without duplicating business logic.

Z-Graph | *Interactive Math Visualization Platform* | z-graph.vercel.app | github.com/sk-izsk/z-graph

- Built a multi-mode graphing platform in React 18 / TypeScript covering 17 mathematical categories – algebra, calculus, polar, parametric, vector fields, Fourier series, Julia/Mandelbrot fractals, and 3D implicit surfaces – with live formula editing, parameter sliders, animation controls, and a preset navigation system.
- Combined 2D rendering via Mafs and 3D scene rendering via Three.js / React Three Fiber under a shared Zustand state layer; mathjs handles live formula parsing; Web Workers offload fractal sampling and surface meshing off the main thread to keep the UI responsive.
- Added PNG export, fullscreen mode, i18n, theme switching, and formula explanation cards, all wired through Zod-validated React Hook Form inputs and a category-based config structure that keeps adding new graph types low-friction.

ZTable Mobile | *React Native Chemistry Reference App* | github.com/sk-izsk/rn-z-table

- Refactored a React Native / Expo chemistry app into a feature-based architecture: thin Expo Router route shells in `app/`, dedicated domain modules in `features/chemistry` (formula parsing, equation balancing, molar mass), `features/elements`, `features/table`, and `features/tools` (worksheet builder, solubility lookup, ion reference).
- Extracted oversized components into focused hooks and sub-components, split large locale dictionaries into structured i18next modules, and brought typecheck, oxlint, oxfmt, and the full Bun test suite to a passing state without changing any public-facing behavior.
- 12 passing tests across 5 chemistry-focused test files covering formula parsing, unicode subscript normalization, equation balancing, molar mass calculation, worksheet generation, and solubility mapping, all enforced in the build path.

ZTable Web | *Interactive Chemistry App* | zsk-periodic.vercel.app | github.com/sk-izsk/zsk-periodic

- Designed a performance-focused render pipeline: equation balancing runs off the main thread in a Web Worker using nullspace-based fraction math, route screens lazy-load through TanStack Router, and 118 periodic table cells use memoized Zustand selectors to prevent broad re-renders on interaction.
- Built with React 19, TypeScript, Vite, and Bun across 8 features including 3D atom visualization (Three.js / React Three Fiber), molar mass parsing, solubility lookup, PDF worksheet export (jspdf), and i18n – with a typed JSON data pipeline, codegen scripts, and full Vitest + oxlint + typecheck coverage enforced in the build path.

Productify | *Full-Stack Product Management App* | izsk.dev | github.com/sk-izsk

- Built a full-stack React 19 and Node.js application using Vite, React Router 7, and compound component pattern for a scalable, maintainable UI.
- Architected a strongly-typed REST API using Express 5, Drizzle ORM, and PostgreSQL (Neon) for relational product and comment data.
- Integrated TanStack Query for cache-aware mutations and Clerk SDK for end-to-end authentication with route protection and role-based authorization.

EDUCATION

Concordia University

Montréal, QC

M.Eng., Electrical & Computer Engineering

Sept 2016 – Dec 2018

- Relevant coursework: Computer Architecture Design, Microprocessor Applications, Digital Signal Processing, Model-Driven Software Engineering, Formal Hardware Verification, Real-Time Applications, Robotics.

DecodeMTL

Montréal, QC

Full-Stack Web Development Bootcamp

April 2019 – May 2019

- Intensive full-stack web development program. Technologies: JavaScript, React, Node.js, Express, MongoDB, HTML5, CSS3.